

Azure Data Factory Project(data Flow)

1. Pipeline

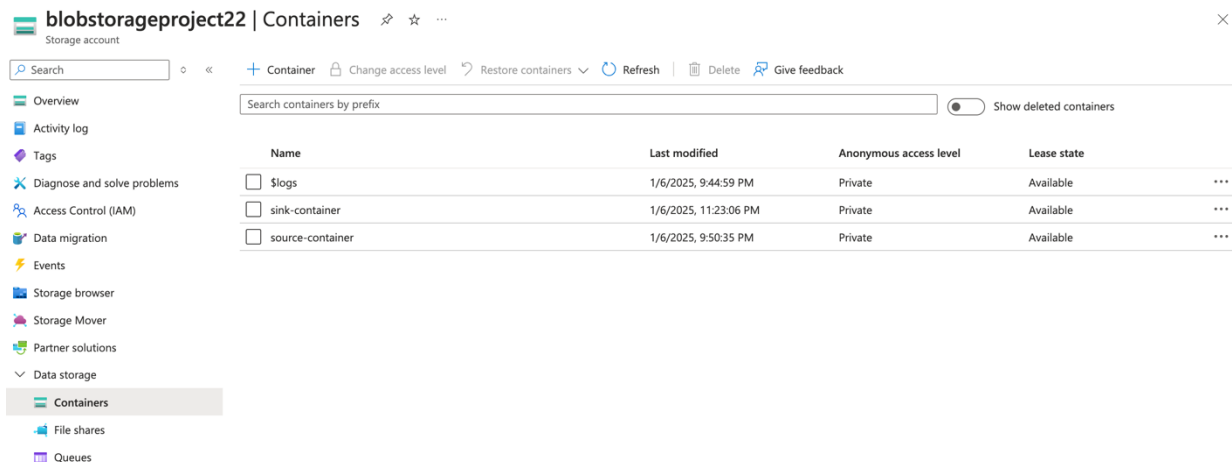
Requirements:

- Use mapping data flow to transform sales the dataset.xlsx
- Filter revenue greater than 2000
- Find out the age group with highest revenue profit and profit margin
- Store the transformed data in csv.

Design:

- Metadata validation.
- Data transformations via Mapping Data Flow.

1. Create a blob storage and containers:



blobstorageproject22 | Containers

Storage account

Search

Container Change access level Restore containers Refresh Delete Give feedback

Search containers by prefix

Show deleted containers

Name	Last modified	Anonymous access level	Lease state
☐ \$logs	1/6/2025, 9:44:59 PM	Private	Available
☐ sink-container	1/6/2025, 11:23:06 PM	Private	Available
☐ source-container	1/6/2025, 9:50:35 PM	Private	Available

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Data storage

Containers

File shares

Queues

Source container

source-container

Container

Search

UploadChange access levelRefreshDeleteChange tierAcquire leaseBreak leaseView snapshotsCreate snapshotGive feedback

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: source-container

Search blobs by prefix (case-sensitive)

Show deleted blobs

Add filter

Name	Modified	Access tier	Archive status	Blob type	Size	Lease state
bike_sales_data.xlsx	1/6/2025, 11:16:21 PM	Hot (Inferred)		Block blob	26.94 KiB	Available

2. Create a ADF instance:

2.1 create a linked service

Data Factory

Validate all

Publish all

Preview experience

Off

General

Factory settings

Connections

Linked services

Integration runtimes

Microsoft Purview

Source control

Git configuration

ARM template

Author

Triggers

Global parameters

Data flow libraries

Linked services

Linked service defines the connection information to a data store or compute. Learn more

New

Filter by name

Annotations : Any

Showing 1 - 2 of 2 items

Name	Type	Related	Annotations
sink_LS	Azure Blob Storage	1	
source_LS	Azure Blob Storage	1	

2.2 create a datasets

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Search factory and documentation

ashishcloud1@outlook.com

Factory Resources

- Pipelines: 1
 - pipeline1
- Change Data Capture (preview): 0
- Datasets: 2
 - bike_sink_Datasets
 - bike_source_Dataset
- Data flows: 1
 - dataflow
- Power Query: 0

bike_source_Dataset

Excel bike_source_Dataset

Connection Schema Parameters

Linked service: source_LS

File path: source-container / Directory / bike_sales_data.xlsx

Compression type: No compression

Worksheet mode: Name (selected), Index

Sheet name: Bike Sales

Range: e.g. A3:H5

Null value:

First row as header: ☒

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Search factory and documentation

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Factory Resources

- Pipelines: 1
 - pipeline1
- Change Data Capture (preview): 0
- Datasets: 2
 - bike_sink_dataset
 - bike_source_Dataset
- Data flows: 1
 - dataflow
- Power Query: 0

bike_sink_dataset

DelimitedText bike_sink_dataset

CSV

Connection Schema Parameters

Linked service: sink_LS

File path: sink-container / Directory / File name

Compression type: No compression

Column delimiter: Comma (,)

Row delimiter: Default (\r\n, or \n)

Encoding: Default(UTF-8)

Quote character: Double quote (")

Escape character: Backslash (\)

2.3 Add Pipeline Activities

1. Validate Source File

Activity: Get Metadata:

The screenshot shows the Azure Data Factory interface. On the left, the 'Factory Resources' pane lists 'Pipelines' (1), 'Data flows' (1), and 'Power Query' (0). The 'Activities' pane shows a search bar and a list of activity categories. The main canvas displays a 'dataflow' activity named 'dataflow' with a 'Get Metadata' activity (Get Metadata1) connected to an 'If Condition' activity (If Condition1). The 'If Condition' activity has two paths: 'True' leading to 'Data flow1' and 'False' leading to a 'Wait' activity. The 'Settings' tab for the 'Get Metadata' activity is open, showing the 'Dataset' as 'bike_source_Dataset' and the 'Field list' with 'Argument', 'Exists', and 'Last modified' selected. The 'Start time (UTC)' and 'End time (UTC)' fields are also visible.

Activity: if condition:

- True: Continue with processing with data flow.
- False: wait for 10 sec than fail activity

The screenshot shows the Azure Data Factory interface with the 'If Condition' activity (If Condition1) selected. The main canvas displays the 'If Condition' activity with two paths: 'True' leading to 'Data flow1' and 'False' leading to a 'Wait1' activity followed by a 'Fail1' activity. The 'Settings' tab for the 'If Condition' activity is open, showing the 'Expression' as '@activity("Get Metadata1").output.exi...'. The 'Activities (3)' tab is also visible, showing the 'Data flow1' activity and the 'Wait1' and 'Fail1' activities.

Activity: mapping dataflow



1. Source Transformation

Source settings Source options Projection Optimize **Inspect** Data preview ●

Output stream name * [Learn more](#)

Description

Import data from bike_source_Dataset

[Reset](#)

Source type *

Dataset

Inline

Dataset *

bike_source_Dataset

[Test connection](#) [Open](#) [New](#)

Options

☒ Allow schema drift ⓘ
☐ Infer drifted column types ⓘ
☐ Validate schema ⓘ

Microsoft Azure | Data Factory | datafactoryproj22 | Search factory and documentation

» Data Factory | Validate all | Publish all | Preview experience | Off

» bike_source_Dataset | bike_sink_dataset | pipeline1 | dataflow

✓ Validate | Data flow debug | Debug Settings

Number of rows: INSERT 89 | UPDATE 0 | DELETE 0 | UPSERT 0

Refresh | Typecast | Modify | Map drifted | Statistics | Remove | Export to CSV

	Sales_Or...	Date	Day	Month	Year	Custom...	Age_Gro...	Custom...	Country	State
+	261695	2021-12...	1	December	2021	39	Adults (3...	F	United S...	Calif
+	261695	2021-12...	1	December	2021	44	Adults (3...	M	United K...	Engl
+	261697	2021-12...	2	December	2021	37	Adults (3...	M	United S...	Calif
+	261698	2021-12...	2	December	2021	31	Young A...	F	Australia	New
+	261699	2021-12...	3	December	2021	37	Adults (3...	F	United S...	Calif
+	261700	2021-12...	3	December	2021	24	Youth (<...	F	United K...	Engl

2. Select Transformation:

The screenshot shows the 'select' transformation settings in the Databricks Data Factory. The pipeline is named 'pipeline1' and is in the 'dataflow' state. The 'select' transformation is highlighted, showing its configuration:

- Output stream name:** select
- Description:** remove unnecessary columns
- Incoming stream:** source
- Options:**
 - ☒ Skip duplicate input columns
 - ☒ Skip duplicate output columns
- Input columns:** 12 mappings: 5 column(s) from the incoming stream. The mapping table shows 'Sales_Order #' mapped to 'Sales_Order_id'.

The pipeline diagram above the settings shows the flow from 'source' to 'select' (12 total columns), then to 'derivedColumn' (derived column profit margin%), 'filter1' (Filtering rows using expressions on columns 'Revenue'), 'aggregate1' (Aggregating data by 'Customer_Gender', 'Age_Group' producing columns 'Order_Quantity'), 'sort1' (Sorting rows on columns 'Total_Profit', 'Revenue'), and finally to 'sink1' (Export data to bike_sink_dataset).

The screenshot shows the 'Data preview' of the 'select' transformation. The pipeline is named 'pipeline1' and is in the 'dataflow' state. The 'Data preview' tab is selected, showing the following data:

Number of rows	INSERT	UPDATE	DELETE	UPSERT	LOOKUP
89	0	0	0	0	0

The data preview table shows the following columns: Sales_Order, Date, Age_Group, Customer, Country, Product_Cat, Sub_Cate, Product_D, and Order_Qu. The data is sorted by Sales_Order in descending order.

Sales_Order	Date	Age_Group	Customer	Country	Product_Cat	Sub_Cate	Product_D	Order_Qu
261695	2021-12-01	Adults (35...)	F	United Sta...	Bikes	Mountain ...	Mountain...	4
261695	2021-12-01	Adults (35...)	M	United Kin...	Bikes	Mountain ...	Mountain...	1
261697	2021-12-02	Adults (35...)	M	United Sta...	Bikes	Mountain ...	Mountain...	2
261698	2021-12-02	Young Ad...	F	Australia	Bikes	Mountain ...	Mountain...	1
261699	2021-12-03	Adults (35...)	F	United Sta...	Bikes	Mountain ...	Mountain...	2
261700	2021-12-03	Youth (<25)	F	United Kin...	Bikes	Mountain ...	Mountain...	1
261701	2021-12-03	Adults (35...)	M	United Sta...	Bikes	Mountain ...	Mountain...	1

3. Derived Column Transformations:

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» Data Factory | Validate all | Publish all | Preview experience

» bike_source_Dataset | bike_sink_dataset | pipeline1 | dataflow

✓ Validate | Data flow debug | Debug Settings

Reference: 1
Columns: 17 total
source

select
remove unnecessary columns

derivedColumn
Columns: 15 total

filter1
Filtering rows using expressions on columns 'Revenue'

aggregate1
Aggregating data by 'Customer_Gender, Age_Group' producing columns 'Order_Quantity, Total_Profit, Revenue'

sort1
Sorting rows on columns 'Total_Profit, Revenue'

sink1
Export data to bike_sink_dataset

Derived column's settings | Optimize | Inspect | Data preview

Output stream name *
derivedColumn
Learn more

Description
derived column profit margin%

Incoming stream *
select

+ Add | Clone | Delete | Open expression builder

Columns * 0

Column	Expression
Total_Cost	{ Unit_Cost } * Order_Quantity
Total_Profit	((Unit_Price) - { Unit_Cost }) * Order_Quantity
Profit_Margin	round((((Unit_Price) - { Unit_Cost }) * Order_Quantity

Data Factory | Validate all | Publish all | Preview experience Off

» bike_source_Dataset | bike_sink_dataset | pipeline1 | dataflow

✓ Validate | Data flow debug | Debug Settings

source
Import data from bike_source_Dataset

select
remove unnecessary columns

derivedColumn
Columns: 15 total

filter1
Filtering rows using expressions on columns 'Revenue'

aggregate1
Aggregating data by 'Customer_Gender, Age_Group' producing columns 'Order_Quantity, Total_Profit, Revenue'

sort1
Sorting rows on columns 'Total_Profit, Revenue'

sink1
Export data to bike_sink_dataset

Derived column's settings | Optimize | Inspect | Data preview

0 | UPSERT 0 | LOOKUP 0 | ERROR 0 | TOTAL 89

Sub_Cate...	Product...	Order_Qu...	Unit_Cost	Unit_Price	Revenue	Total_cost	Total_Profit	Profit_Ma...
Mountain ...	Mountain...	4	1252	2295	9180	5008	4172	45.44
Mountain ...	Mountain...	1	1266	2320	2320	1266	1054	45.43
Mountain ...	Mountain...	2	420	769	1538	840	698	45.38
Mountain ...	Mountain...	1	420	769	769	420	349	45.38
Mountain ...	Mountain...	2	0	2295	4590	0	4590	100.0
Mountain ...	Mountain...	1	1252	2295	2295	1252	1043	45.44
Mountain ...	Mountain...	1	1252	2295	2295	1252	1043	45.44

4. Filter Transformation:

Data Factory

Validate all

Publish all

Preview exp

bike_source_Dataset

bike_sink_dataset

pipeline1

dataflow

Validate

Data flow debug

Debug Settings

source

Import data from bike_source_Dataset

+

select

remove unnecessary columns

+

derivedColumn

derived column profit margin%

+

filter1

Columns: 15 total

+

aggregate1

Aggregating data by 'Customer_Gender, Age_Group' producing columns 'Order_Quantity, 'Total_Profit, Revenue'

+

sort1

Sorting rows on columns 'Total_Profit, Revenue'

Filter settings

Optimize

Inspect

Data preview

Output stream name *

filter1

Learn more

Description

Filtering rows using expressions on columns 'Revenue'

Reset

Incoming stream *

derivedColumn

Filter on *

Revenue > 2000.

Filter settings

Optimize

Inspect

Data preview

Number of rows

+ INSERT

74

* UPDATE

0

Refresh

Typecast

Modify

Map drifted

Statistics

Re

	↑↓ Sales_Ord...	123	↑↓ Date	12	↑↓ Age_Group	abc	↑↓ Customer...
+	261695		2021-12-01		Adults (35...		F
+	261695		2021-12-01		Adults (35...		M

5. Aggregate Transformation:

The screenshot shows the configuration of an 'aggregate1' transformation in Apache NiFi. The top section displays a workflow diagram with the following steps: 'source' (Import data from bike_source_Dataset), 'select' (remove unnecessary columns), 'derivedColumn' (derived column profit margin%), 'filter1' (Filtering rows using expressions on columns 'Revenue'), 'aggregate1' (Columns: 5 total), and 'sort1' (Sorting rows on columns Total_Profit, Revenue). Below the diagram, the 'Aggregate settings' tab is active. It shows the 'Output stream name' as 'aggregate1', a description of aggregating data by 'Customer_Gender, Age_Group' producing columns 'Order_Quantity, Total_Profit, Revenue', and the 'Incoming stream' as 'filter1'. The 'Group by' section is set to 'Aggregates'. The 'Columns' section lists 'Customer_Gender' and 'Age_Group' with their respective 'Name as' values. A 'Reset' button is also visible.

Incoming stream * filter1

Group by Aggregates

Grouped by: Customer_Gender, Age_Group

[+ Add](#) [Clone](#) [Delete](#) [Open expression builder](#)

☐	Column	Expression
☐	Order_Quantity	count(Order_Quantity) 12L + 🗑
☐	Total_Profit	sum(Total_Profit) 12L + 🗑
☐	Revenue	sum(Revenue) 12L + 🗑



Aggregate settings

Optimize

Inspect

Data preview

Number of rows

+

INSERT

6

+

UPDATE

0

✗

DELETE

0

+

UPSERT

0

🔍

LOOKUP

0

✗

ERROR

0

↻

Refresh

▼

Typecast

▼

🔗

Modify

▼

📄

Map drifted

📊

Statistics

✕

Remove

📄

Export to CSV

▼

📄

↑↓	Customer_Gender	abc	↑↓	Age_Group	abc	↑↓	Order_Quantity	12L	↑↓	Total_Profit	12L	↑↓	Revenue	12L	↑↓
+	F			Adults (35-64)			22			53444			112600		
+	M			Adults (35-64)			20			41107			90461		
+	F			Youth (<25)			5			9342			20560		
+	M			Youth (<25)			2			5263			11587		
+	M			Young Adults (25-34)			10			17074			37576		
+	F			Young Adults (25-34)			15			33465			74527		

6. Sort Transformation:

bike_source_Dataset bike_sink_dataset pipeline1 dataflow

Validate
 Data flow debug
 Debug Settings

source
Import data from bike_source_Dataset
 +
 select
remove unnecessary columns
 +
 derivedColumn
derived column profit margin%
 +
 filter1
Filtering rows using expressions on columns 'Revenue'
 +
 aggregate1
Aggregating data by 'Customer_Gender, Age_Group' producing columns 'Order_Quantity,'
 +
 sort1
Columns: 5 total
 +
 sink
Export bike_si...

Sort settings Optimize Inspect Data preview ●

Description

Sorting rows on columns 'Total_Profit, Revenue'

Reset

Incoming stream *

aggregate1

Options *

☐ Case insensitive

☐ Sort only within partition

Sort conditions *

aggregate1's column	Order	Nulls first
121 Total_Profit	Ascending	☒
121 Revenue	Descending	☒

Data Factory Validate all Publish all Preview experience

bike_source_Dataset bike_sink_dataset pipeline1 dataflow

Validate
 Data flow debug
 Debug Settings

source
Import data from bike_source_Dataset
 +
 select
remove unnecessary columns
 +
 derivedColumn
derived column profit margin%
 +
 filter1
Filtering rows using expressions on columns 'Revenue'
 +
 aggregate1
Aggregating data by 'Customer_Gender, Age_Group' producing columns 'Order_Quantity,'
 +
 sort1
Columns: 5 total
 +
 sink1
Export dati...

Sort settings Optimize Inspect Data preview ●

Number of rows + INSERT 6 UPDATE 0 DELETE 0 UPSERT 0 LOOKUP 0 ERROR 0

Refresh

Typecast

Modify

Map drifted

Statistics

Remove

Export to CSV

	↑↓ Customer_Gender	abc ↑↓ Age_Group	abc ↑↓ Order_Quantity	121 ↑↓ Total_Profit	121 ↑↓ Revenue
+	M	Youth (<25)	2	5263	11587
+	F	Youth (<25)	5	9342	20560
+	M	Young Adults (25-34)	10	17074	37576
+	F	Young Adults (25-34)	15	33465	74527
+	M	Adults (35-64)	20	41107	90461
+	F	Adults (35-64)	22	53444	112600

7. Sink Transformation:

Home > sink-container >

sink-container

Container

Search

Upload

Change access level

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: sink-container

Search blobs by prefix (case-...)

Show deleted blobs

Add filter

Name

☐ _SUCCESS

☐ part-00000-40201073-25b5-4f70-8f9d-1b22fba8385d-c000

part-00000-40201073-25b5-4f70-8f9d-1b22fba8385d-c000.csv

Blob

Save Discard Download Refresh Delete

Overview

Versions

Snapshots

Edit

Generate SAS

1 Customer_Gender, Age_Group, Order_Quantity, Total_Profit, Revenue

2 M, Youth (<25), 2, 5263, 11587

3 F, Youth (<25), 5, 9342, 20560

4 M, Young Adults (25-34), 10, 17074, 37576

5 F, Young Adults (25-34), 15, 33465, 74527

6 M, Adults (35-64), 20, 41107, 90461

7 F, Adults (35-64), 22, 53444, 112600

8

Csv

Preview